
Unit – 2 : Session – 3 : SLO – 2 | Interprocess Communication — Task Defender

Scenario 1: shared counter incremented by P1 and P2

Communication Type: Shared memory (both processes use the shared variable counter).
Synchronization Needed: Yes, mutual exclusion around the increment (mutex or semaphore).
Possible Problem: Race condition, lost updates and an inconsistent counter value.

Scenario 2: Process A sends an image to Process B over a network

Communication Type: Message passing (send and receive over the network, for example sockets).
Synchronization Needed: Yes, blocking (synchronous) send with an acknowledgement from B.
Possible Problem: A can block forever if the acknowledgement or message is lost, so a timeout and retry is needed.

Scenario 3: producer writes to a shared buffer, consumer reads from it

Communication Type: Shared memory (a bounded buffer shared by producer and consumer).
Synchronization Needed: Yes, semaphores (empty and full counts) plus a mutex on the buffer.
Possible Problem: Buffer overflow or underflow and race conditions on the buffer; busy waiting or deadlock if the waits are ordered wrongly.

Scenario 4: threads update a shared account balance

Communication Type: Shared memory (threads of one process share the same data).
Synchronization Needed: Yes, a mutex lock or an atomic update around each balance change.
Possible Problem: Race condition, so the final balance is wrong (lost update).